

4.3 Dynamic Characteristics of a Measurement System

- Only a few measurements are constant or slowly varying quantities. Most instruments must process signals that are functions of time. It is this time-varying property of signals that requires us to consider dynamic instrument characteristics.
- Differential or integral equations are required to relate dynamic inputs to dynamic outputs for continuous systems.

只有少数被测量是恒值或缓慢变化的，大多数情况下还需考虑系统的动态特性；

微积分方程将连续系统的动态输入与输出联系起来。

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.1 Mathematical Description of Dynamic Characteristics

- Ordinary Linear Differential Equations
微分方程
- Transfer Function
传递函数
- Frequency Response Function
频率响应函数
- Unit-step Response Function
阶跃响应函数
- Unit-Impulse Response Function
单位脉冲响应函数

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Ordinary Linear Differential Equations 微分方程

Fortunately, many engineering instruments can be described by *ordinary linear differential equations* with constant coefficients. The input $x(t)$ is related to the output $y(t)$ according to the following equation:

$$a_n \frac{d^n y}{dt^n} + \dots + a_1 \frac{dy}{dt} + a_0 y(t) = b_m \frac{d^m x}{dt^m} + \dots + b_1 \frac{dx}{dt} + b_0 x(t)$$

$$(a_n D^n + \dots + a_1 D + a_0) y(t) = (b_m D^m + \dots + b_1 D + b_0) x(t)$$


Readers familiar with Laplace transforms may recognize that D can be replaced by the Laplace parameter s to obtain the equation relating the transforms $Y(s)$ and $X(s)$. This is a linear differential equation and the coefficients are not functions of time or the input $x(t)$.

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Transfer Function 传递函数

Frequency-domain transfer function is the ratio of the Laplace transform $Y(s)$ of the output $y(t)$ to the Laplace transform $X(s)$ of the input $x(t)$, with its initial conditions and equilibrium point to be zero, described as $H(s)$.

$$H(s) = \frac{Y(s)}{X(s)} \quad \text{where } s \text{ is a complex variable } \sigma + j\omega \text{ and } s \text{ is called the Laplace operator. (拉氏变换算子)}$$

The Laplace transform of this equation, with zero initial conditions, is

$$(a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0) Y(s) = (b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0) X(s)$$

Hence, the frequency transfer function of a measuring system is

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_m s^m + b_{m-1} s^{m-1} + \dots + b_1 s + b_0}{a_n s^n + a_{n-1} s^{n-1} + \dots + a_1 s + a_0}$$

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• Features of the transfer function:

- The transfer function depends on the measuring system itself, independent of the input signals. $H(s)$ 与输入信号 $x(t)$ 及系统的初始状态无关, 系统的动态特性完全由 $H(s)$ 决定。
- Different physical systems may have identical transfer functions. $H(s)$ 只反映系统传输特性, 和系统具体物理结构无关。同一形式的传递函数可表征具有相同传输特性的不同物理系统, 可以是电路系统, 也可以是机械系统。
- The denominator of $H(s)$ depends on the structure of the system; the numerator is the relationship between the system and the outside world, such as the position of the input (excitation) point, the input method, the measured and measuring point layout. $H(s)$ 的分母取决于系统的结构 (分母中 s 的幂次 n 代表系统微分方程的阶数); 分子则是系统与外界之间的关系, 如输入 (激励) 点的位置、输入方式、被测量及测点布置情况有关。
- The transfer function is completely equivalent to the differential equation and can be transformed into each other. 传递函数与微分方程完全等价, 可以相互转化。

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Frequency Response Function 频率响应函数

Definition: The frequency response function is defined as the ratio of the Fourier transform $Y(j\omega)$ of the output $y(t)$ to the Fourier transform $X(j\omega)$ of the input $x(t)$, with its initial conditions and equilibrium point to be zero, described as $H(j\omega)$ or $H(\omega)$.

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

The frequency response function of a measuring system, with all initial conditions zero, is

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{b_m(j\omega)^m + b_{m-1}(j\omega)^{m-1} + \cdots + b_1(j\omega) + b_0}{a_n(j\omega)^n + a_{n-1}(j\omega)^{n-1} + \cdots + a_1(j\omega) + a_0} \quad (m \leq n)$$

The frequency response function can be obtained by substituting $j\omega$ for s or letting $s = j\omega$. Therefore, the frequency response function is a particular case of transfer function.

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Frequency Response Function 频率响应函数

Representation: Frequency response $H(j\omega)$ is a function that relates the output response to a sinusoidal input at frequency ω . In general, it is a complex with real and imaginary parts. We can separate it into its magnitude (amplitude response) and its phase angle (phase response).

$$H(j\omega) = P(\omega) + jQ(\omega) = |H(j\omega)|e^{j\angle H(j\omega)} = A(\omega)e^{j\varphi(\omega)}$$

where $A(\omega) = |H(j\omega)| = \sqrt{P(\omega)^2 + Q(\omega)^2}$ is called magnitude which is the amplitude ratio of the output and input sinusoids, and $\varphi(\omega) = \angle H(j\omega) = \arctan \frac{Q(\omega)}{P(\omega)}$ is the phase angle which is the phase shift between the input and output sinusoids.

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Frequency Response Function 频率响应函数

Frequency Retention Property

$$\text{if } x(t) = X \sin \omega t = X e^{j\omega t}$$

$$\text{then } y(t) = Y \sin(\omega t + \varphi) = Y e^{j(\omega t + \varphi)}$$

$$H(j\omega) = \frac{b_m(j\omega)^m + b_{m-1}(j\omega)^{m-1} + \dots + b_1(j\omega) + b_0}{a_n(j\omega)^n + a_{n-1}(j\omega)^{n-1} + \dots + a_1(j\omega) + a_0} = \frac{Y(\omega)}{X(\omega)} = \frac{Y}{X} e^{j\varphi(\omega)}$$

The frequency transfer function is a complex quantity having a magnitude that is the ratio of the magnitude of the output to the magnitude of the input and a phase angle that is the phase of the output $y(t)$ minus the phase of the input $x(t)$.

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

- Graphical description of frequency response characteristics 频率响应特性的图形描述:

1) The plots of $A(\omega) - \omega$ magnitude frequency and $\varphi(\omega) - \omega$ phase frequency spectrum:

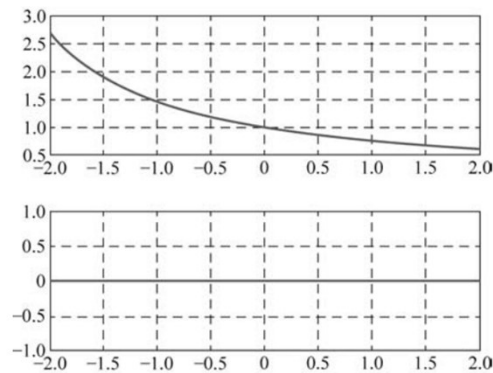


Figure 4.5 Magnitude and phase frequency spectrum of first-order system

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

- Graphical description of frequency response characteristics 频率响应特性的图形描述:

2) Using logarithmic scales, a pair of graphs $20\log A(\omega) - \log \omega$ and $\varphi(\omega) - \log \omega$ is referred to as a Bode plot or Bode diagram, also named Logarithmic separate magnitude and phase frequency spectrum, shown as Figure 4.6:

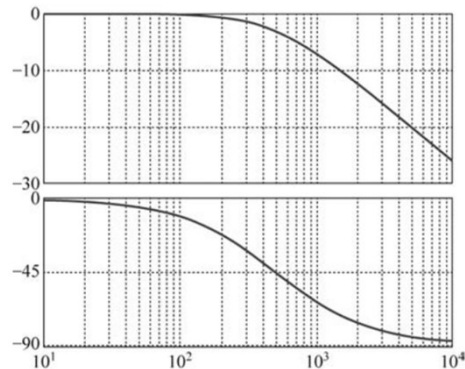


Figure 4.6 Bode diagram

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- Graphical description of frequency response characteristics 频率响应特性的图形描述:

3) We can use a polar plot to represent frequency response. The graph is the real part against the imaginary part of frequency response, that is $P(\omega) - Q(\omega)$, referred to as a Nyquist plot, shown as Figure 4.7:

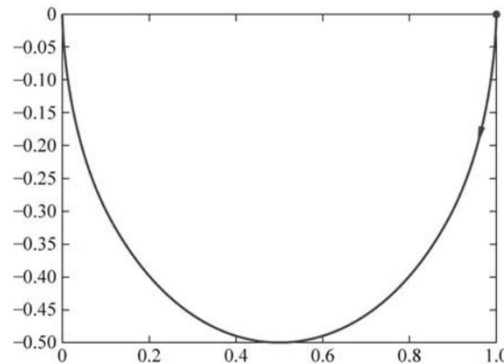


Figure 4.7 Nyquist plot

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

- How to find the frequency response function: 频率响应函数的求法

Theoretical Computation

A cosine function input $x(t) = X_0 \cos \omega_0 t$

let $X_0 = 1$, By Euler formula $e^{\pm j\omega t} = \cos(\omega t) \pm j \sin(\omega t)$, The cosine function input can be also represented as

$$x(t) = \frac{1}{2}(e^{j\omega_0 t} + e^{-j\omega_0 t})$$

if the input is $e^{j\omega_0 t}$ and $h(t)$ is the impulse response, then according to convolution and the output is

$$h(t) * e^{j\omega_0 t} = \int_{-\infty}^{+\infty} h(\tau) e^{j\omega_0(t-\tau)} d\tau = e^{j\omega_0 t} \int_{-\infty}^{+\infty} h(\tau) e^{-j\omega_0 \tau} d\tau = H(j\omega_0) \cdot e^{j\omega_0 t}$$

Also, if the input is $e^{-j\omega_0 t}$, we can get the output is $H(-j\omega_0) \cdot e^{-j\omega_0 t}$ which is conjugate complex of $H(j\omega_0) \cdot e^{j\omega_0 t}$.

$$y(t) = \frac{1}{2}[H(j\omega_0)e^{j\omega_0 t} + H(-j\omega_0)e^{-j\omega_0 t}]$$

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

- How to find the frequency response function: 频率响应函数的求法

Theoretical Computation

$$y(t) = \frac{1}{2} [H(j\omega_0)e^{j\omega_0 t} + H(-j\omega_0)e^{-j\omega_0 t}] \quad y(t) = \operatorname{Re}\{H(j\omega_0)e^{j\omega_0 t}\}$$

$$y(t) = \operatorname{Re}\{|H(j\omega_0)|e^{j\angle H(j\omega_0)}e^{j\omega_0 t}\} \quad y(t) = |H(j\omega_0)|\operatorname{Re}\{e^{j\omega_0 t + j\angle H(j\omega_0)}\}$$

$$y(t) = |H(j\omega_0)| \cos(\omega_0 t + \angle H(j\omega_0))$$

where $|H(j\omega_0)|$ is the magnitude of the system, and $\angle H(j\omega_0)$ is the phase shift of the system.

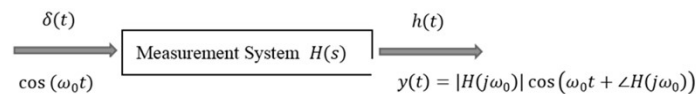


Figure 4.8 Frequency Response

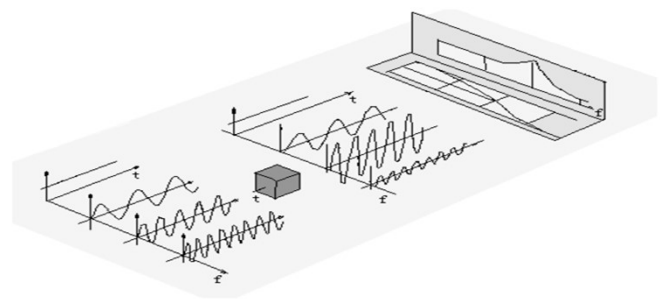
4.3 Dynamic Characteristics of a Measurement System

■ 4.3.1 Mathematical Description of Dynamic Characteristics

- How to find the frequency response function: 频率响应函数的求法

Frequency Response Tests

A sinusoidal signal is input to the system, and the output is measured in steady state. Therefore, we can get the amplitude of input X_i , the amplitude of output Y_i and the phase shift φ_i . Repeating this process with different frequencies changed $\Delta\omega$ each time. A set of Y_i/X_i and φ_i can be drawn in the frequency response plot. The frequency response is obtained. 依次用不同频率 f_i 的简谐信号去激励被测系统, 同时测出激励和系统的稳态输出的幅值、相位, 得到幅值比 A_i 、相位差 φ_i 。



从系统最低测量频率 f_{min} 到最高测量频率 f_{max} , 逐步增加正弦激励信号频率 f , 记录下各频率对应的幅值比和相位差, 绘图就得到系统幅频和相频特性。

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■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Unit-step Response Function

If the input of the system is a unit step signal, then

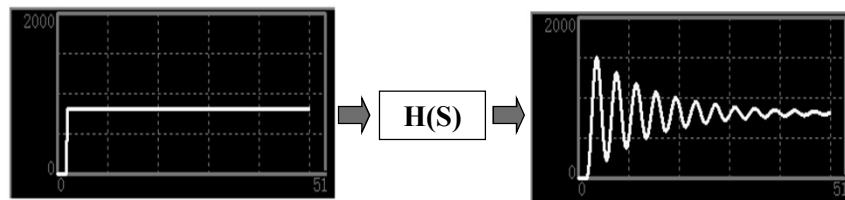
$$X(s) = \mathcal{L}[x(t)] = 1/s$$

The Laplace transform of the output signal is

$$Y(s) = H(s) \cdot X(s) = H(s)/s$$

Through the inverse Laplace transform of $Y(s)$, the step response signal $y(t)$ can be obtained.

The shape of the response curve can help indicate the order of a system. We can know whether the system is underdamped, overdamped or critically damped especially in second-order system.



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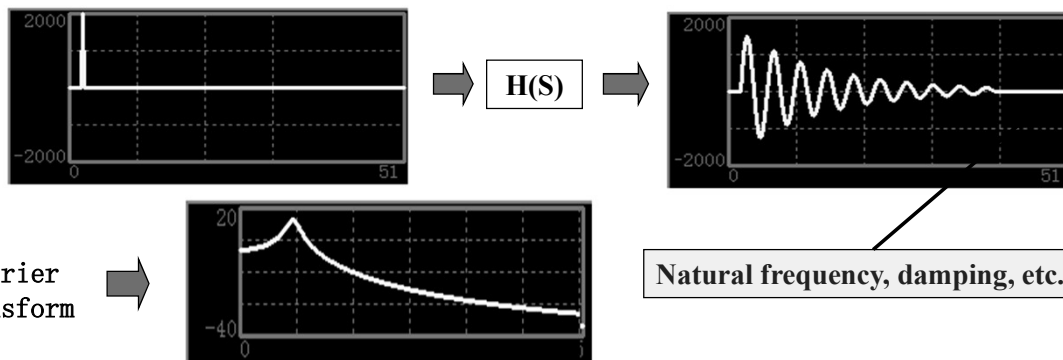
■ 4.3.1 Mathematical Description of Dynamic Characteristics

• Unit-Impulse Response Function

If the input of the system is unit pulse $\delta(t)$, since the Laplace transform of $\delta(t)$ is 1, there is:

$$Y(s) = H(s), \text{ then } y(t) = \mathcal{L}^{-1}[H(s)] = h(t)$$

$h(t)$ is called the impulse response function

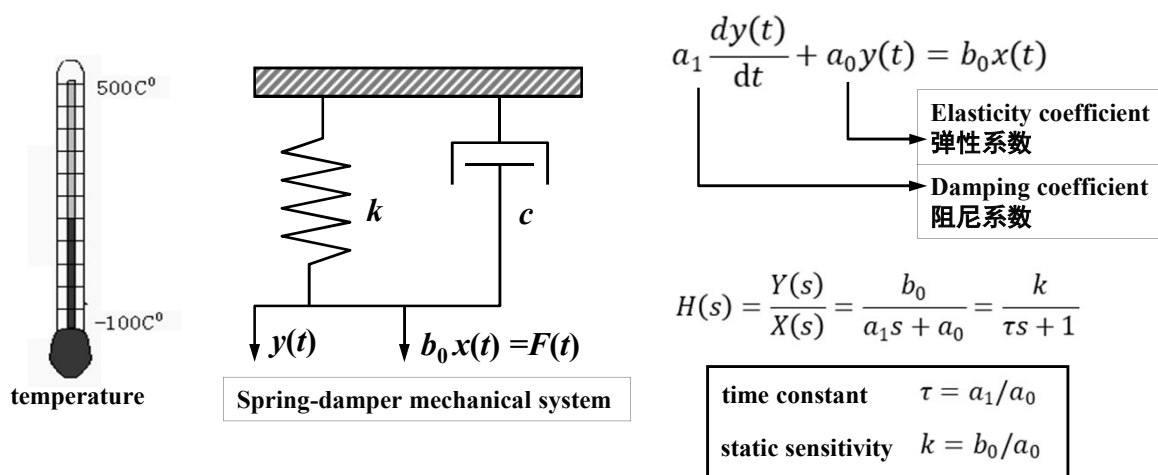


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- Mathematical models are useful to characterize the performance of a measurement system and predict how it responds to specific inputs.
- Any higher order system can be regarded as the series or parallel connection of some first-order and second-order systems.
- Therefore, analyzing and understanding the transmission characteristics of the first-order and second-order measurement systems is the basis of analyzing and understanding the transmission characteristics of high-order complex systems.

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.2 First-order Measurement System



4.3 Dynamic Characteristics of a Measurement System

■ 4.3.2 First-order Measurement System

■ Example 4.3

A passive filter in measurement system can be regarded as a simple electronic circuit with a resistor R and a capacitor C shown in Figure 4.9. Determine the transfer function of this passive filter.

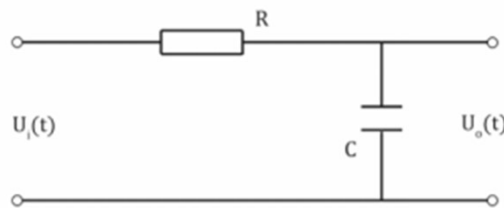


Figure 4.9 Passive Filter

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.2 First-order Measurement System

Solution: By Kirchhoff's current law

$$\begin{cases} u_i(t) = i(t)R + \frac{1}{C} \int i(t) dt \\ u_o(t) = \frac{1}{C} \int i(t) dt \end{cases}$$

Take Laplace transform

$$\begin{cases} U_i(s) = I(s)R + \frac{1}{Cs} I(s) \\ U_o(s) = \frac{1}{Cs} I(s) \end{cases}$$

So the transfer function of this passive filter is

$$H(s) = \frac{U_o(s)}{U_i(s)} = \frac{\frac{1}{Cs}}{R + \frac{1}{Cs}} = \frac{1}{RCs + 1} = \frac{1}{\tau s + 1}$$

where $\tau = RC$ is the time constant of the passive filter.

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.2 First-order Measurement System

• Unit-step Response of First-order System

For a first-order system subject to a unit-step input, the Laplace transform of unit-step function is $1/s$, we get the output in s domain (letting $k=1$ for simplicity) is

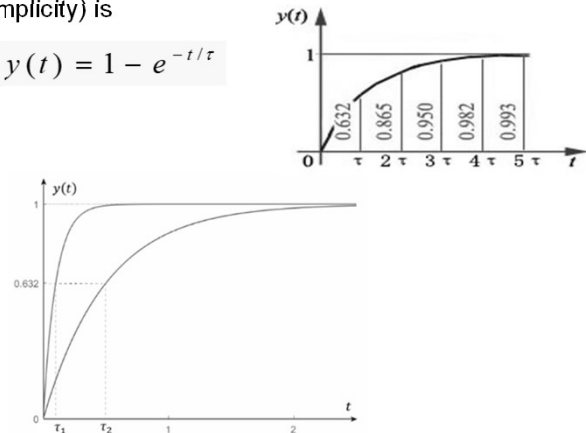
$$Y(s) = H(s)X(s) = \frac{1}{\tau s + 1} \cdot \frac{1}{s} \implies y(t) = 1 - e^{-t/\tau}$$

$t = \tau, y(t) = 0.632$.

The smaller the time constant τ , the faster the system response.

The time constant is an important dynamic characteristic of first-order system shown in Figure 4.10.

When $t \geq 4\tau$, the response reaches 98.2%. In engineering, we usually assume that the first-order system has reached its steady-state value within four time constants.



4.3 Dynamic Characteristics of a Measurement System

■ 4.3.2 First-order Measurement System

• Frequency Response of First-order System

We can obtain frequency response of the first-order system by substituting $j\omega$ for s in $H(s)$ as

$$H(s) = \frac{Y(s)}{X(s)} = \frac{b_0}{a_1 s + a_0} = \frac{k}{\tau s + 1} \implies H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = \frac{1}{j\tau\omega + 1}$$

The amplitude ratio of the output to the input is

$$A(j\omega) = |H(j\omega)| = \frac{1}{\sqrt{1 + (\tau\omega)^2}}$$

The phase shift of the output with the input is

$$\varphi(\omega) = -\arctan(\tau\omega)$$

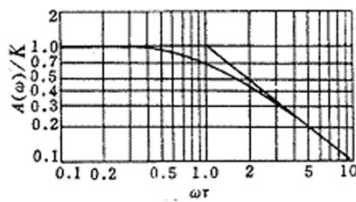
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■ 4.3.2 First-order Measurement System

- Frequency Response of First-order System

$$A(j\omega) = |H(j\omega)| = \frac{1}{\sqrt{1 + (\tau\omega)^2}}$$

bode diagram



$$\varphi(\omega) = -\arctan(\tau\omega)$$

Amplitude-frequency
and phase-frequency
curves

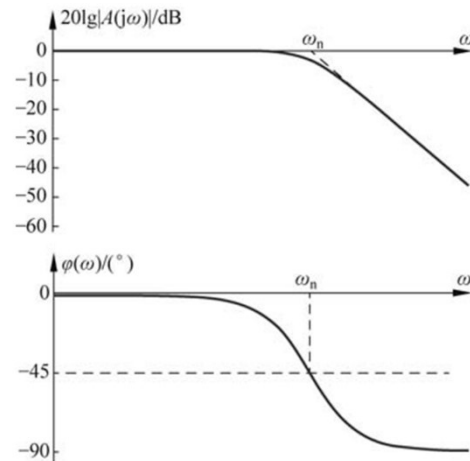
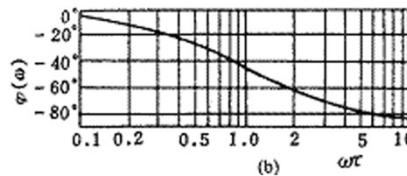


Figure 4.11 Frequency response of first-order system

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.2 First-order Measurement System

- For sinusoids, the magnitude of the output decreases as frequency increases. For larger time constants, this decrease occurs at lower frequency.
- When $\omega = 1/\tau$, the magnitude is 0.707 times smaller, and the phase angle is -45° . This particular frequency is known as the corner, cutoff, or break frequency (转角频率、截断频率).
- This is a low-pass filter(低通滤波器); low-frequency sinusoids are not severely attenuated, whereas high-frequency sinusoids produce very little output voltage.

◆幅频特性 $A(\omega)$ 和相频特性 $\varphi(\omega)$ 表示输入和输出之间的差异，称为稳态响应动态误差。

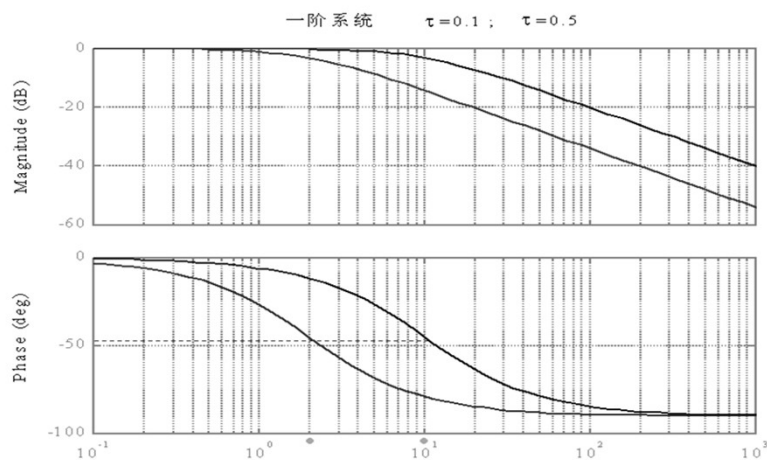
实际应用中常限定幅值误差

$$\varepsilon = \left| \frac{A(\omega) - A(0)}{A(0)} \right| \times 100\% = |A(\omega) - 1| \times 100\% \leq \text{某个给定值}$$

◆一阶系统的特性：

一低通性质：幅值比 $A(\omega)$ 随输入频率 ω 的增大而减小。

一系统的工作频率范围取决于时间常数 τ 。当 $\omega\tau$ 较小时，幅值和相位的失真都较小。当 $\omega\tau$ 一定时， τ 越小，测试系统的工作频率范围越宽。



◆一阶系统适用于测量缓变或低频被测量；为了减小系统的动态误差，增大工作频率范围，应尽可能采用时间常数小的测试系统。

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4.3.2 First-order Measurement System

- A first-order low-pass instrument has a time constant of 20 ms. Find the maximal sinusoidal input frequency that will keep output error due to frequency response less than 5%. Find the phase angle at this frequency.

Solution:

$$\frac{Y(j\omega)}{X(j\omega)} = \frac{K}{1+j\omega\tau}$$

$$\left| \frac{K}{1+j\omega\tau} \right| = \frac{K}{\sqrt{1+\omega^2\tau^2}} = 0.95K$$

$$(\omega^2\tau^2 + 1)(0.95)^2 = 1$$

$$\omega^2 = \frac{1 - (0.95)^2}{(0.95)^2 (20 \times 10^{-3})^2}$$

$$\omega = 16.4 \text{ rad/s}$$

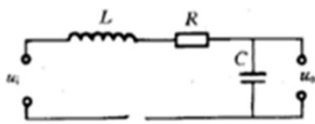
$$f = \frac{\omega}{2\pi} = 2.62 \text{ Hz}$$

$$\phi = \tan^{-1} \left(\frac{-\omega\tau}{1} \right) = -18.2^\circ$$

4.3 Dynamic Characteristics of a Measurement System

4.3.3 Second-order Measurement System

- An instrument is second order if a second-order differential equation is required to describe its dynamic response.



$$a_2 \frac{d^2 y(t)}{dt^2} + a_1 \frac{dy(t)}{dt} + a_0 y(t) = b_0 x(t)$$

$$\omega_n = \sqrt{\frac{a_0}{a_2}}$$

$$\xi = \frac{a_1}{2\sqrt{a_0 a_2}}$$

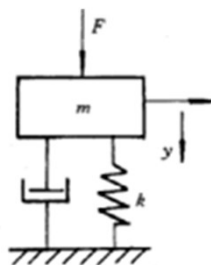
$$S = \frac{b_0}{a_0}$$

$$\frac{d^2 y(t)}{dt^2} + 2\xi\omega_n \frac{dy(t)}{dt} + \omega_n^2 y(t) = S\omega_n^2 x(t)$$



$$H(s) = \frac{\omega_n^2}{s^2 + 2\xi\omega_n s + \omega_n^2}$$

$$H(j\omega) = \frac{1}{[1 - (\frac{\omega}{\omega_n})^2] + 2j\xi \frac{\omega}{\omega_n}}$$

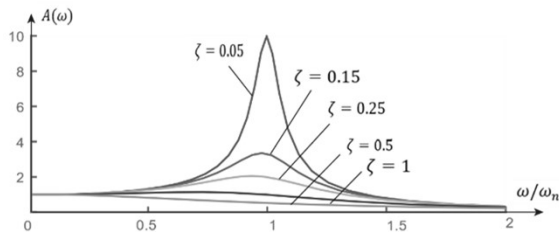


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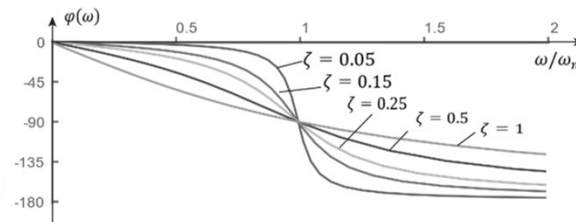
■ 4.3.3 Second-order Measurement System

• Frequency Response of Second-order System

$$H(j\omega) = \frac{1}{[1 - (\frac{\omega}{\omega_n})^2] + 2j\xi \frac{\omega}{\omega_n}}$$



$$A(\omega) = \frac{1}{\sqrt{[1 - (\frac{\omega}{\omega_n})^2]^2 + 4\xi^2 (\frac{\omega}{\omega_n})^2}}$$



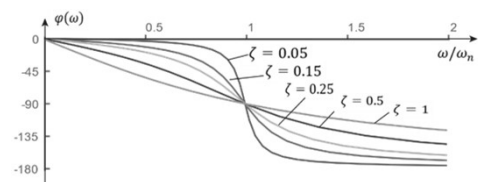
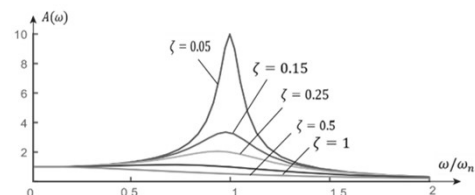
$$\varphi(\omega) = -\arctg \frac{2\xi (\frac{\omega}{\omega_n})}{1 - (\frac{\omega}{\omega_n})^2}$$

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.3 Second-order Measurement System

• Frequency Response of Second-order System

- When $\omega/\omega_n = 1$, resonance occurs. Therefore, ω_n is also named the resonant frequency. For small damping ratios, the output amplitude becomes quite large at the resonant frequency.
- For input frequencies less than the natural frequency, $\omega/\omega_n < 1$, the amplitude ratio is close to 1. And for $\omega/\omega_n > 1$, the amplitude ratio is close to 0.
- The phase angle is negative, which means the output lags behind the input.
- For $\omega/\omega_n \ll 1$, $\tan \varphi$ is approximately 0 and $\varphi = 0^\circ$. And for $\omega/\omega_n \gg 1$, $\tan \varphi$ is approximately $-\infty$ and $\varphi = -180^\circ$. And for $\omega/\omega_n = 1$, $\tan \varphi$ is approximately $-\infty$ and $\varphi = -90^\circ$.



◆二阶系统的特性：

—低通特性

—频率响应与阻尼比 ζ 有关

- 1) 当 $\zeta > 0.707$ 时, $A(\omega) \leq 1$, 无谐振, $A(\omega)$ 随 ω 增加而单调下降。
- 2) 当 $\zeta < 0.707$ 时, 在 $\frac{\omega}{\omega_0} \approx 1$ 处 (谐振频率处) 产生谐振, $A(\omega)$ 有峰值。
- 3) 当 $\zeta = 0$, 在 $\omega_r = \omega_0$ 处, $A(\omega) \rightarrow \infty$

$$\text{由 } \frac{dA(\omega)}{d\omega} = 0$$

$$\text{得谐振频率 } \omega_r = \omega_0 \sqrt{1 - 2\zeta^2} \quad (0 \leq \zeta \leq 0.707)$$

$$A(\omega)_{\max} = \frac{1}{2\zeta \sqrt{1 - \zeta^2}}$$

—频率响应与固有频率 ω_0 有关：固有频率越高，保持动态误差在一定范围内的工作频率范围越宽，反之越窄。

对二阶系统通常推荐采用阻尼比 $\zeta = 0.7$ 左右，且可用频率在 $0 \sim 0.6\omega_0$ 范围内变化，测试系统可获得较好的动态特性，其幅值误差不超过5%，同时相频特性接近于直线，即测试系统的动态特性误差较小。

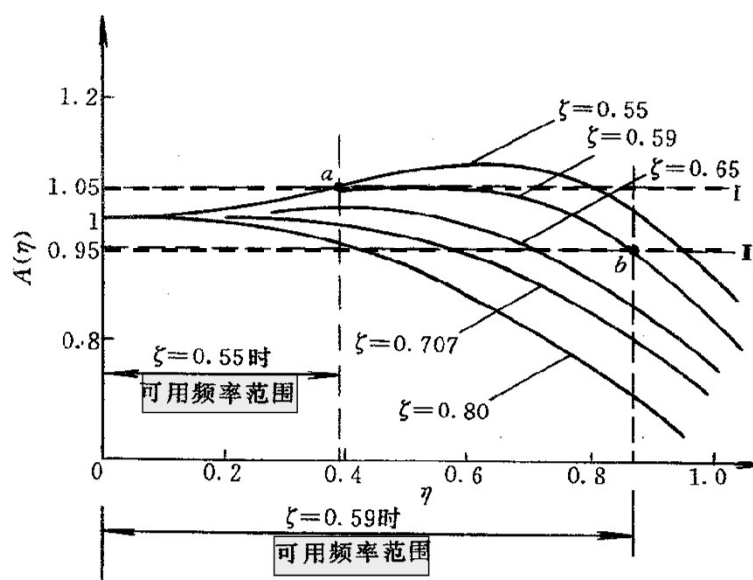
在动态测试时，必须了解测试系统的可用频率范围——与允许的幅值误差 ε 和阻尼比 ζ 有关。

——允许的幅值误差 ε 越小，其可用频率范围越窄；反之，其可用频率范围越宽。

—— $\zeta = 0.7$ 左右时，也有较宽的可用频率范围。

选择、设计测试系统时尤为重要！

不同阻尼比对可用频率范围的影响



4.3 Dynamic Characteristics of a Measurement System

■ 4.3.3 Second-order Measurement System

• Unit-step Response of Second-order System

$$H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

Case 1: Overdamped case ($\zeta > 1$): Unit-step response of the second-order system is

$$y(t) = 1 - \frac{(\zeta + \sqrt{\zeta^2 - 1}) e^{(-\zeta + \sqrt{\zeta^2 - 1})\omega_n t}}{2\sqrt{\zeta^2 - 1}} + \frac{(\zeta - \sqrt{\zeta^2 - 1}) e^{-(\zeta + \sqrt{\zeta^2 - 1})\omega_n t}}{2\sqrt{\zeta^2 - 1}}.$$

Case 2: Critically damped case ($\zeta = 1$): Unit-step response of the second-order system is

$$y(t) = 1 - (1 + \omega_n t) e^{-\omega_n t}.$$

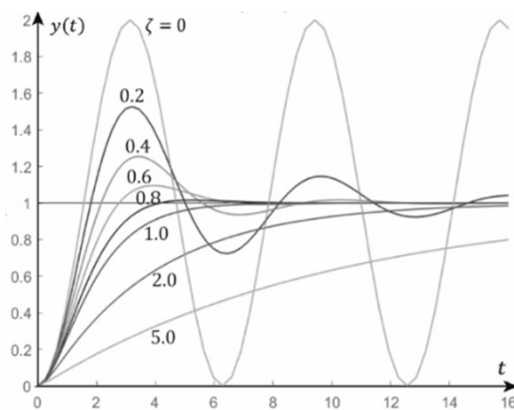
Case 3: Underdamped case ($0 < \zeta < 1$): Unit-step response of the second-order system is

$$y(t) = 1 - \frac{e^{-\zeta\omega_n t}}{\sqrt{1 - \zeta^2}} \sin(\omega_d t + \varphi), \quad \omega_d = \omega_n \sqrt{1 - \zeta^2} \quad \varphi = \arctan \frac{\sqrt{1 - \zeta^2}}{\zeta}.$$

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.3 Second-order Measurement System

• Unit-step Response of Second-order System



- An under damped system has a fast rise time, but it overshoots and may take a while to settle.
- A critically damped system has an average rise time with no overshoot and settles the most rapidly.
- And an overdamped system has no overshoot but has a slow rise time.
- The response speed is related to the natural frequency. When the damping rate is constant, the higher the natural frequency ω_0 , the faster the response speed, and vice versa.

二阶系统阶跃响应的特性：

- ◆ ζ 值过大或过小，趋于最终稳态值的时间都过长。通常取 $\zeta = 0.6 \sim 0.8$ ，响应速度快，动态误差小，系统的输出才能以较快的速度达到给定的误差范围。
- ◆ 响应速度与固有频率有关。阻尼率一定时，固有频率 ω_0 越高，响应速度越快，反之越慢。

4.3 Dynamic Characteristics of a Measurement System

4.3.3 Second-order Measurement System

Unit-step Response of Second-order System

For commonly used second-order test systems, usually take $\xi < 1$

$$y(t) = 1 - \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \sin(\omega_d t + \varphi),$$

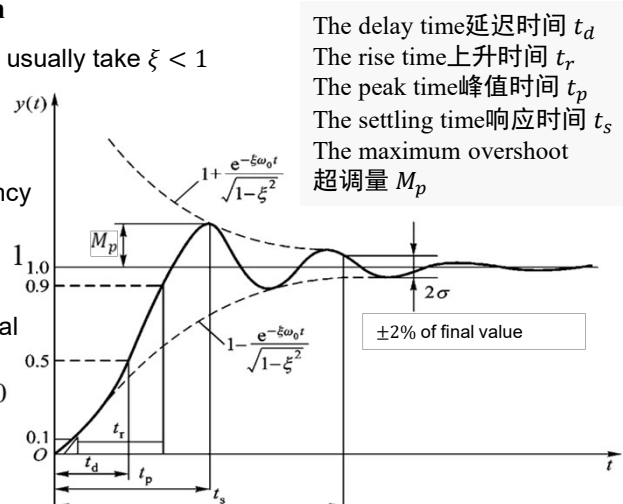
$$\omega_d = \omega_n \sqrt{1-\xi^2} \quad \text{damped oscillation frequency}$$

$$\varphi = \arctan \frac{\sqrt{1-\xi^2}}{\xi}. \quad \text{when } \xi = 0 \quad \frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \rightarrow 1$$

The system produces an undecayed sinusoidal oscillation with ω_n

when $\xi \neq 0$, as t increases to ∞ , $\frac{e^{-\xi\omega_n t}}{\sqrt{1-\xi^2}} \rightarrow 0$

The system produces ω_d damped oscillation



4.3 Dynamic Characteristics of a Measurement System

4.3.3 Second-order Measurement System

Unit-step Response of Second-order System

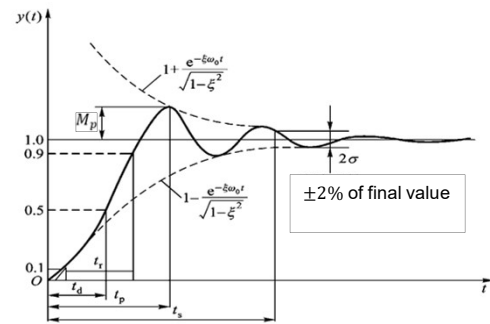
The delay time t_d is the time required for the response to reach of the final value, the very first time.

The rise time t_r is the time required for the signal to go from 0.1 to 0.9 of its steady-state value. Let $\omega_d = \omega_n \sqrt{1 - \xi^2}$ and $\beta = \arctan \frac{\sqrt{1 - \xi^2}}{\xi}$. Then $t_r = \frac{\pi - \beta}{\omega_d}$.

The peak time t_p is the time at which the first peak value occurs. $t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \xi^2}}$

The settling time t_s is the time required for a signal to stay within $\pm 2\%$ of its final value for all future times, simplified as $t_s \approx \frac{4}{\xi \omega_n}$.

The maximum overshoot M_p is the maximum amount of the output exceeds the steady state value, usually specified as a percentage of the steady state value.



$$M_p = \frac{x(t_p) - x_0(\infty)}{x_0(\infty)} \times 100\% = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\%$$

4.3 Dynamic Characteristics of a Measurement System

4.3.3 Second-order Measurement System

Example 4.7

A transfer function of second-order system is shown as $H(s) = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$ where $\zeta = 0.6$ and $\omega_n = 5 \text{ rad/s}$. Determine the rise time t_r , peak time t_p , maximum overshoot M_p , and settling time t_s when the system is subjected to a unit-step input.

Solution: For the given values of ζ and ω_n , we can obtain

Rise time t_r :

$$t_r = \frac{\pi - \beta}{\omega_d} = \frac{\pi - \arctan \frac{\sqrt{1 - \zeta^2}}{\zeta}}{\omega_n \sqrt{1 - \zeta^2}} = \frac{3.14 - 0.93}{4} = 0.55 \text{ s}$$

Peak time t_p :

$$t_p = \frac{\pi}{\omega_d} = \frac{\pi}{\omega_n \sqrt{1 - \zeta^2}} = \frac{3.14}{4} = 0.785 \text{ s}$$

Maximum overshoot M_p :

$$M_p = e^{\frac{-\zeta\pi}{\sqrt{1-\zeta^2}}} \times 100\% = e^{\frac{-0.6 \times 3.14}{\sqrt{1-0.6^2}}} \times 100\% = 9.5\%$$

Settling time t_s for the $\pm 2\%$ criterion:

$$t_s \approx \frac{4}{\zeta \omega_n} = \frac{4}{3} = 1.33 \text{ s}$$

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.3 Second-order Measurement System

• Example 4.8

The elastic sensor can convert a force input F into a displacement output x shown in Figure 4.16. There are a mass m kg, a spring of stiffness k N · m⁻¹ and a damper of constant c N · s · m⁻¹ with initial conditions zero. Determine the transfer function of this elastic sensor.

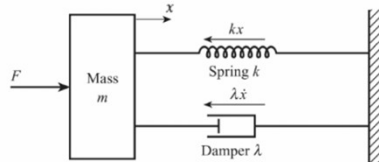


Figure 4.16 Mass-spring-damper Model of Elastic Sensor

Solution: By Newton's second law

$$F - kx - c \frac{dx}{dt} = m \frac{d^2x}{dt^2}$$

Then, a second-order linear differential equation can be obtained based on zero initial conditions

$$\frac{m}{k} \frac{d^2x}{dt^2} + \frac{c}{k} \frac{dx}{dt} + x = \frac{1}{k} F$$

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.3 Second-order Measurement System

• Example 4.8

Take Laplace transform, we can get

$$\frac{m}{k} s^2 x(s) + \frac{c}{k} s x(s) + x(s) = \frac{1}{k} F(s)$$

and the transfer function is

$$\frac{x(s)}{F(s)} = \frac{\frac{1}{k}}{\frac{m}{k} s^2 + \frac{c}{k} s + 1} = \frac{1}{ms^2 + cs + k}$$

And if undamped natural frequency is

$$\omega_n = \sqrt{\frac{k}{m}} \text{ rad/s}$$

damping ratio is

$$\zeta = \frac{c}{2\sqrt{km}}$$

We can obtain the mathematical model of this elastic sensor as

$$H(s) = \frac{x(s)}{F(s)} = \frac{\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$$

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.4 Calibration of Dynamic Characteristics of Systems

- A new measuring instrument should be calibrated by an instrument manufacturer to ensure that it will initially behave according to the static and dynamic characteristics stated in the specifications. After repaired, modified or a specified usage, if the characteristics of the instrument have drifted from the standard specifications unacceptably, an instrument has to be recalibrated back to standard specifications.
- We use tests to determine the system response to some input which is an alternative way of developing a mathematical model for a real system. Two ways are often used, frequency response method(频率响应法) and step response method(阶跃响应法).

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.4 Calibration of Dynamic Characteristics of Systems

• First-order System

For first-order instruments, there is only one parameter related to dynamic response, the time constant τ , which may be found by different methods.

(1) Step response method

The step response function of the first-order system is

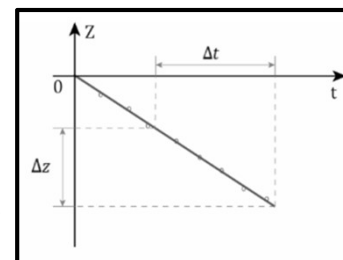
$$y(t) = 1 - e^{-t/\tau}$$

then

$$1 - y(t) = e^{-t/\tau}$$

We define $z = \ln[1 - y(t)]$, and then

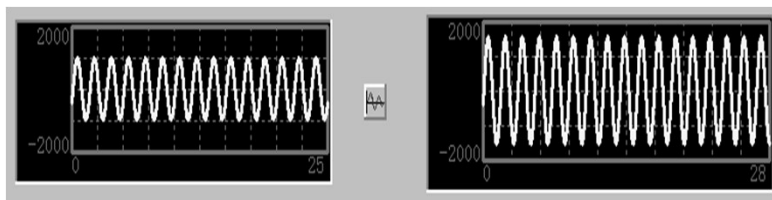
$$z = -\frac{t}{\tau}, \quad \frac{\Delta z}{\Delta t} = -\frac{1}{\tau}$$



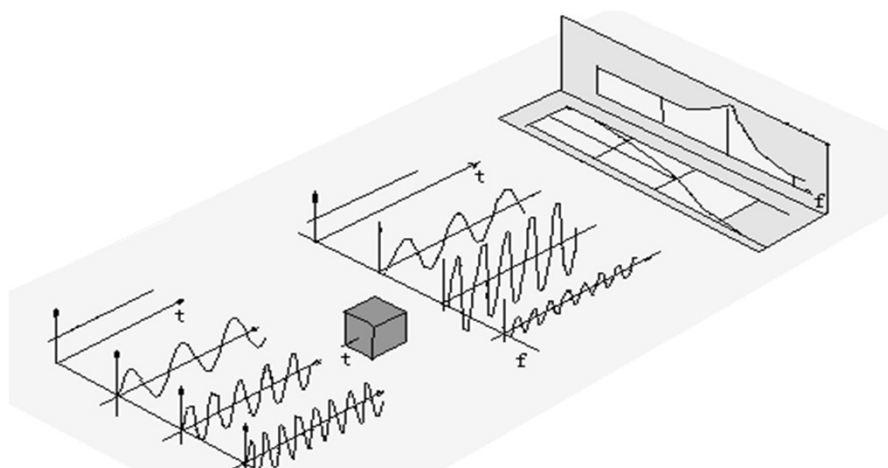
Then according to the slope of the curve, the time constant τ is known

3) 实验法——正弦激励法

依次用不同频率 f_i 的简谐信号去激励被测系统，同时测出激励和系统的稳态输出的幅值、相位，得到幅值比 A_i 、相位差 ϕ_i 。



频率响应函数是描述系统的简谐输入和其稳态输出的关系，在求解系统频率响应函数时，必须在系统响应达到稳态阶段时才测量。



从系统最低测量频率 f_{min} 到最高测量频率 f_{max} ，逐步增加正弦激励信号频率 f ，记录下各频率对应的幅值比和相位差，绘图就得到系统幅频和相频特性。

4.3 Dynamic Characteristics of a Measurement System

■ 4.3.4 Calibration of Dynamic Characteristics of Systems

• First-order System

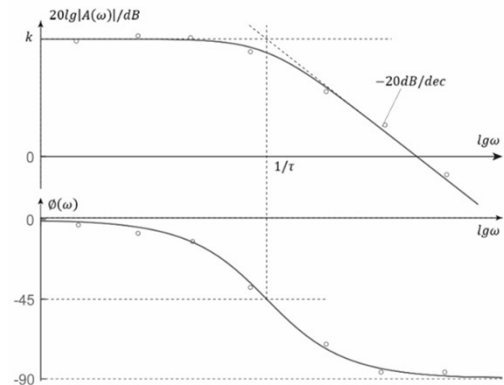
(2) Frequency response method

After testing, amplitude ratio and phase angle are plotted on the logarithmic scales. From Equations

$$A(\omega) = |H(j\omega)| = \frac{1}{\sqrt{1 + (\omega\tau)^2}}$$

$$\phi(\omega) = \angle H(j\omega) = -\arctan(\omega\tau)$$

when $\omega\tau = 1$, the amplitude ratio $A(\omega) = 1/\sqrt{2}$ and phase difference $\phi = -45^\circ$. The time constant $\tau = 1/\omega$ at breakpoint as dynamic characteristics first-order system is found from these results as listed in right figure.



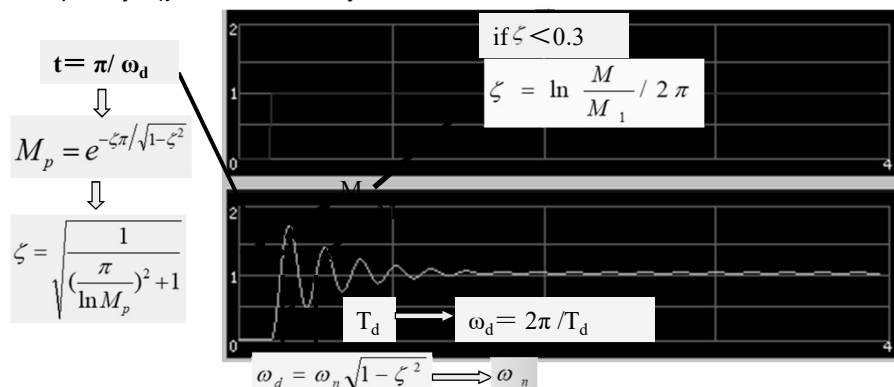
4.3 Dynamic Characteristics of a Measurement System

■ 4.3.4 Calibration of Dynamic Characteristics of Systems

• Second-order System

(1) Step response method

For an under damped ($\zeta < 1$) second-order system, the value of damping ratio ζ and the value of natural frequency ω_n can be found by



4.3 Dynamic Characteristics of a Measurement System

■ 4.3.4 Calibration of Dynamic Characteristics of Systems

• Second-order System

(2) Frequency response method

For an under damped second-order system, the amplitude ratio of the output and input is given by

$$|A(\omega)|_{\max} = \frac{A_r}{A_o} = \frac{1}{2\zeta\sqrt{1-\zeta^2}} \quad \text{solve for } \zeta$$

Then, the natural frequency ω_n can be found by

$$\omega_r = \omega_n \sqrt{1-2\zeta^2}$$

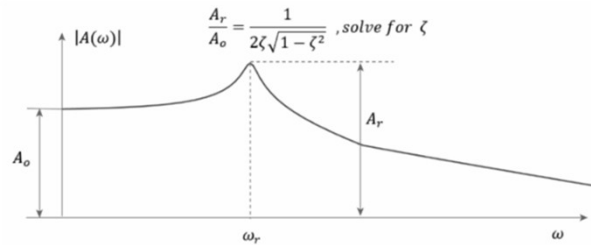


Figure 4.19 Frequency Response Test of Second-order System

4.4 Conditions of Distortionless measurement

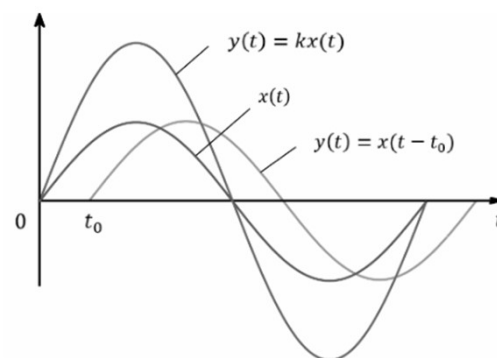
- If $x(t)$ is the input and $y(t)$ is the output, distortionless instrument implies that

$$y(t) = kx(t - t_0) \quad \text{time domain condition}$$

where, k is a constant multiplicative factor and t_0 means a delay in time domain.

• Distortionless measurement features:

A measurement system is said to be distortionless if the input and output have identical wave shapes within a multiplicative constant. A delayed output that returns the input waveform is also considered distortionless.



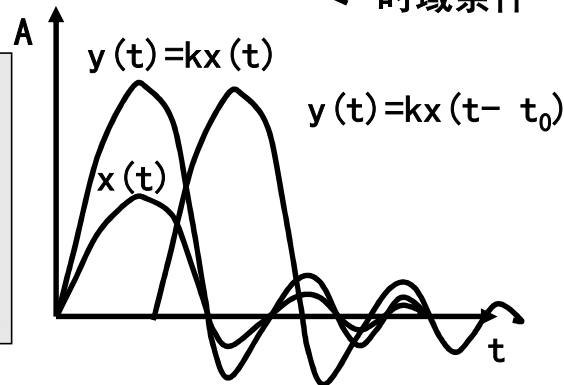
3.4 实现系统不失真测试的条件

设测试系统的输出 $y(t)$ 与输入 $x(t)$ 满足关系

$$y(t) = kx(t - t_0)$$

时域条件

不失真的特性——
系统的输出波形与
输入信号的波形完全
相似，只是幅值放
大了 k 倍，在时间
上延迟了 t_0 而已。



4.4 Conditions of Distortionless measurement

- Taking the Fourier transform

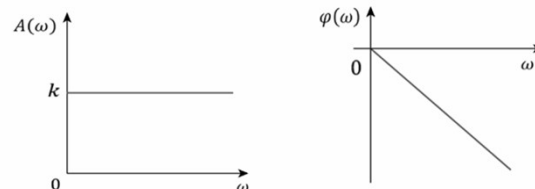
$$y(t) = kx(t - t_0) \longrightarrow Y(j\omega) = kX(j\omega)e^{-j\omega t_0}$$

The transfer function of distortionless system is

Frequency Domain Conditions

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)} = ke^{-j\omega t_0} \longrightarrow A(\omega) = k \quad \varphi(\omega) = -\omega t_0$$

That means amplitude response must be a constant and phase response must be a linear function of ω with slope $-t_0$ which passes through the origin.



进行傅立叶变换

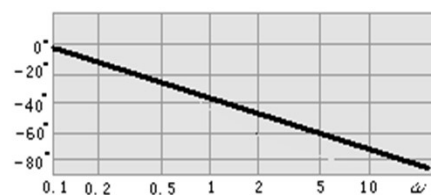
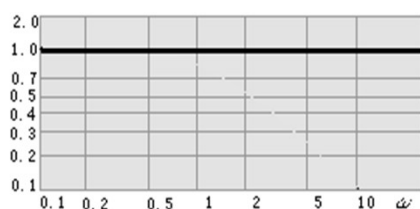
$$y(t)=kx(t-t_0) \longrightarrow Y(j\omega) = kX(j\omega)e^{-j\omega t_0}$$

不失真测试系统的幅频特性和相频特性应分别满足条件

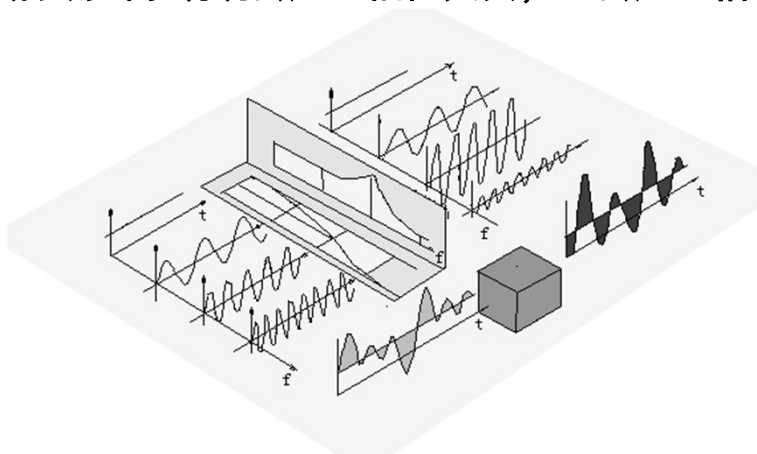
$$A(\omega)=k=\text{常数}$$

$$\phi(\omega)=-\omega t_0$$

频域条件



通常实际测试系统既会产生幅值失真，也会产生相位失真。



只能将波形失真限制在一定的误差范围内。

一阶系统——时间常数越小，则系统的响应越快，近于满足测试不失真条件的频带也越宽。所以一阶系统的时间常数，原则上越小越好。

